

CIRCLES



SYNOPSIS

- **Definition:** The locus of point moving in a plane such that it is at a constant distance from a fixed point is called a circle.

The fixed point is called the centre of the circle and it is denoted by 'C', the constant distance is called its radius and it is denoted by 'r'.

Note: a) If $r = 0$ then the circle is called **point circle**

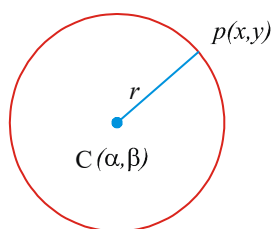
b) If $r = 1$ then the circle is called **unit circle**

c) If $r > 0$ then the circle is called **real circle**

d) If $r < 0$ then the circle is called **imaginary circle**.

- **Equation of a Circle:** The equation of the circle with centre (α, β) and radius 'r' is

$$(x - \alpha)^2 + (y - \beta)^2 = r^2.$$



W.E.-1: Find the equation of the circle with centre $(2, 3)$ and passing through the point $(2, -1)$.

Sol. Let $C = (2, 3)$ and $P = (2, -1)$

Radius of the circle = $CP = 4$

The equation of the required circle is

$$(x - 2)^2 + (y - 3)^2 = 16$$

$$\Rightarrow x^2 + y^2 - 4x - 6y - 3 = 0$$

- **Standard form of the Circle:** The equation of the circle with centre at the origin and radius r is $x^2 + y^2 = r^2$. This is called the standard form of the circle.

- The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a circle if

- $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 \neq 0$
- coefficient of $x^2 =$ coefficient of y^2 ; $(a = b \neq 0)$
- coefficient of xy is zero ; $(h = 0)$
- radius ≥ 0 ; $(g^2 + f^2 - ac \geq 0)$.

- **General form of the Circle:** The equation of the circle of the form $x^2 + y^2 + 2gx + 2fy + c = 0$ is called general equation of the circle and its centre

$$= (-g, -f)$$

$$= \left(\frac{-\text{coefficient of } x}{2}, \frac{-\text{coefficient of } y}{2} \right) \text{ and}$$

$$\text{radius} = \sqrt{g^2 + f^2 - c}.$$

- The general equation of a circle is $ax^2 + ay^2 + 2gx + 2fy + c = 0$ its centre is

$$\left(-\frac{g}{a}, -\frac{f}{a} \right), \text{ radius is } \frac{\sqrt{g^2 + f^2 - ac}}{|a|}$$

W.E.-2: Find the centre and radius of the circle

$$3x^2 + 3y^2 - 6x + 4y - 4 = 0.$$

Sol. The given circle is $x^2 + y^2 - 2x + \frac{4}{3}y - \frac{4}{3} = 0$

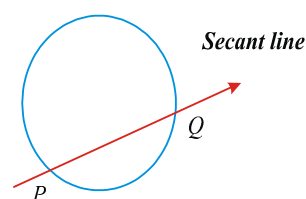
$$\text{Centre} = \left(\frac{-(-2)}{2}, \frac{\frac{-4}{3}}{2} \right) = \left(1, \frac{-2}{3} \right)$$

$$\text{Radius} = \sqrt{g^2 + f^2 - c}$$

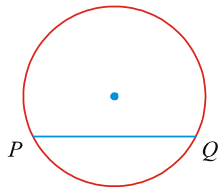
$$= \sqrt{(-1)^2 + \left(\frac{2}{3} \right)^2 + \left(\frac{4}{3} \right)} = \sqrt{\frac{25}{9}} = \frac{5}{3}$$

- Let P and Q be any two points on a circle. Then

- i) The line passing through P and Q is called a secant line.

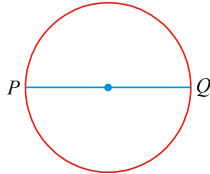


- ii) The line segment joining P and Q is called a chord.



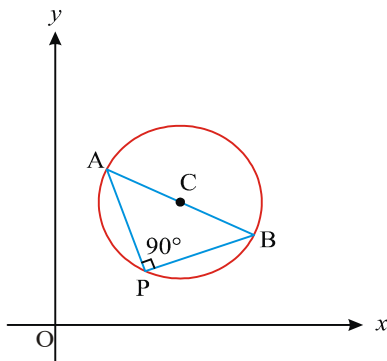
Chord PQ

- iii) A chord passing through centre of a circle is called a diameter of the circle.

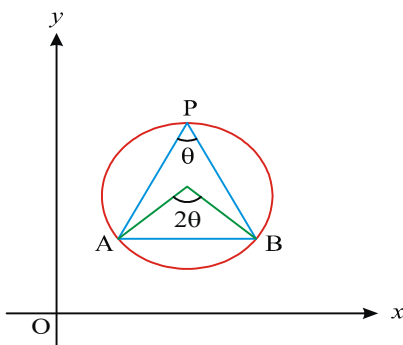


Diameter PQ

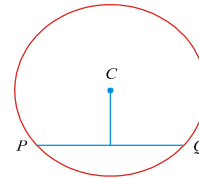
- iv) The diameter of a circle is a chord but a chord need not be a diameter.
v) Every diameter of the circle bisected by the centre of the circle.
vi) Every diameter of the circle cut the circle into two equal parts, each part is called a semi circle.
vii) The angle in a semi circle is right angle.



- viii) The angle made by a chord of a circle at the centre of the circle is double the angle made by the chord at any point on the circumference of the circle lying on the same side of it.



- ix) Perpendicular bisector of any chord of a circle passes through the centre of the circle. **(OR)**
The line joining mid point of a chord and centre of the circle is perpendicular to the chord.



➤ **Equation of a Circle in Diameter form :**

The equation of the circle with (x_1, y_1) and (x_2, y_2) as the extremities of a diameter is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

W.E-3: The equation of the circle which passes through $(1, 0)$ and $(0, 1)$ and has its radius as small as possible is

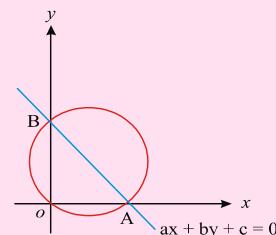
Sol: The radius will be minimum, if the given points are the end points of a diameter. Then the equation of the circle is

$$(x - 1)(x - 0) + (y - 0)(y - 1) = 0$$

$$\Rightarrow x^2 + y^2 - x - y = 0$$

W.E-4: Find the equation of the circum - circle of the triangle formed by the line $ax + by + c = 0$ ($abc \neq 0$) and the coordinate axes.

Sol:



Let the line $ax + by + c = 0$ cut the X,Y axes at A and B respectively

$$\therefore A = \left(-\frac{c}{a}, 0\right) \text{ and } B = \left(0, -\frac{c}{b}\right)$$

We have to find circle passing through A, B and the origin $O(0,0)$. Since $\angle AOB = 90^\circ$

$\therefore$ AB is a diameter of required circle

$\therefore$ Equation of required circle is

$$\left(x + \frac{c}{a}\right)(x - 0) + (y - 0)\left(y + \frac{c}{b}\right) = 0$$

$$\Rightarrow x^2 + y^2 + \frac{c}{a}x + \frac{c}{b}y = 0$$

$$\text{i.e., } \boxed{ab(x^2 + y^2) + c(bx + ay) = 0}$$

- The equation of the circle passing through three points $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ is

$$\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} (x^2 + y^2) + \begin{vmatrix} c_1 & y_1 & 1 \\ c_2 & y_2 & 1 \\ c_3 & y_3 & 1 \end{vmatrix} (x) + \begin{vmatrix} x_1 & c_1 & 1 \\ x_2 & c_2 & 1 \\ x_3 & c_3 & 1 \end{vmatrix} (y) + \begin{vmatrix} x_1 & y_1 & c_1 \\ x_2 & y_2 & c_2 \\ x_3 & y_3 & c_3 \end{vmatrix} = 0$$

$$+ \begin{vmatrix} x_1 & c_1 & 1 \\ x_2 & c_2 & 1 \\ x_3 & c_3 & 1 \end{vmatrix} (y) + \begin{vmatrix} x_1 & y_1 & c_1 \\ x_2 & y_2 & c_2 \\ x_3 & y_3 & c_3 \end{vmatrix} = 0$$

where $c_i = -(x_i^2 + y_i^2)$ for $i=1,2,3$.

- The equation of the circum circle of the triangle formed by the lines $L_1=0, L_2=0, L_3=0$ is $\lambda_1 L_1 L_2 + \lambda_2 L_2 L_3 + \lambda_3 L_3 L_1 = 0$ where $\lambda_1, \lambda_2, \lambda_3$ can be found by using coefficient of $x^2 =$ coefficient of y^2 and coefficient of $xy=0$

W.E-5: Find the equation of the circle which passes through the vertices of the triangle formed by

$$L_1 = x + 3y - 1 = 0, \quad L_2 = x + y + 1 = 0 \quad \text{and} \quad L_3 = 2x + 3y + 4 = 0$$

Sol: Suppose $L_1 L_2; L_2 L_3$ and $L_3 L_1$ intersect at A, B and C respectively. consider a curve whose equation is

$$k(x + 3y - 1)(x + y + 1) + l(x + y + 1)(2x + 3y + 4) + m(2x + 3y + 4)(x + 3y - 1) = 0$$

$$(2x + 3y + 4) + m(2x + 3y + 4)(x + 3y - 1) = 0$$

$$(x + 3y - 1) = 0 \quad \text{--- (1)}$$

i) Coefficient of $x^2 =$ coefficient of y^2

$$k + 2l + 2m = 3k + 3l + 9m$$

$$\text{i.e., } 2k + l + 7m = 0 \quad \text{--- (2)}$$

ii) coefficient of $xy = 0$

$$4k + 5l + 9m = 0 \quad \text{--- (3)}$$

Applying cross multiplication rule for (2) and (3),

$$\begin{array}{ccccc} 1 & \rightarrow & 7 & \rightarrow & 2 \\ 5 & \rightarrow & 9 & \rightarrow & 4 \end{array}$$

we get

$$\frac{k}{9-35} = \frac{l}{28-18} = \frac{m}{10-4}$$

$$\Rightarrow \frac{k}{-26} = \frac{l}{10} = \frac{m}{6} \quad \text{i.e., } \frac{k}{13} = \frac{l}{-5} = \frac{m}{-3}$$

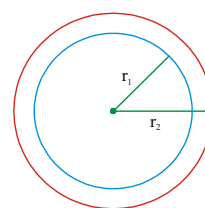
Hence the required equation is

$$13(x + 3y - 1) - 5(x + y + 1) - 3(x + y + 1) = 0$$

$$(2x + 3y + 4) - 3(2x + 3y + 4) - 3(x + 3y - 1) = 0$$

$$\text{i.e., } x^2 + y^2 + 12x + 12y + 7 = 0$$

- **Concentric Circles :** Two or more circles are said to be concentric if they have the same centre and different radii.



- Equation of two concentric circles differ by a constant only. i.e.,

The equation of the circle concentric with the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is of the form $x^2 + y^2 + 2gx + 2fy + k = 0$, k is unknown constant.

W.E- 6: The equation of the circle concentric with the circle $x^2 + y^2 - 6x + 12y + 15 = 0$ and of double its area is (EAMCET-2010)

Sol: From the given circle centre $C = (3, -6)$

$$\text{and radius } r = \sqrt{g^2 + f^2 - c} \Rightarrow r = \sqrt{30}$$

$$A_2 = 2A_1 \Rightarrow \pi r_2^2 = 2\pi r_1^2 \Rightarrow r_2 = \sqrt{2}r_1$$

$$\Rightarrow r_2 = \sqrt{60}$$

The equation of circle is

$$(x - h)^2 + (y - k)^2 = r^2 \Rightarrow (x - 3)^2 + (y + 6)^2 = 60$$

$$\Rightarrow x^2 + y^2 - 6x + 12y - 15 = 0$$

- **Concyclic points :** If the points A, B, C and D lies on the same circle then the points are said to be concyclic points.

- If the lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ meets the co-ordinate axis in four distinct points then those points are concyclic $\Leftrightarrow a_1a_2 = b_1b_2$.

- If the lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ meet the coordinate axes in four distinct concyclic points then the equation of the circle passing through these concyclic points is $(a_1x + b_1y + c_1)(a_2x + b_2y + c_2) - (a_1b_2 + a_2b_1)xy = 0$

W.E-7: If the lines $x - 2y + 3 = 0$, $3x + ky + 7 = 0$ cut the coordinate axes in concyclic points, then k is

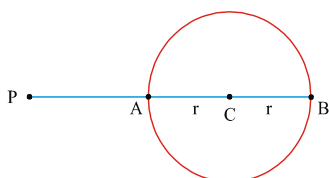
Sol. $a_1a_2 = b_1b_2 \Rightarrow (1)(3) = (-2)(k) \Rightarrow k = \frac{-3}{2}$

W.E-8: If the points $(\sqrt{29}, 0)$, $(5, 2)$, $(2, -5)$, $(-1, k)$ are concyclic ($k \neq 0$) then k is

Sol: Let $A(\sqrt{29}, 0)$, $B(5, 2)$, $C(2, -5)$, $D(-1, k)$

$OA = OB = OC$ and $OB = OD \Rightarrow k = \sqrt{28}$

- **Maximum and minimum distances from a point to the circle :** The minimum distance from the point 'P' to the circle $S = 0$ is $PA = |CP - r|$ and the maximum distance from the point 'P' to the circle $S = 0$ is $PB = |CP + r|$.



W.E-9: The longest distance from $(-3, 2)$ to the circle

$x^2 + y^2 - 2x + 2y + 1 = 0$ is

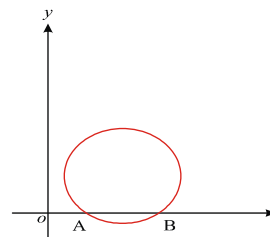
Sol: $x^2 + y^2 - 2x + 2y + 1 = 0$, Centre $C = (1, -1)$,

Radius $r = \sqrt{1+1-1} = 1$

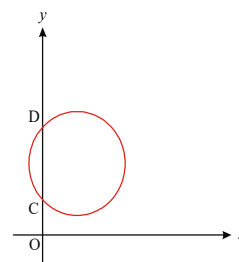
$P = (-3, 2)$

Longest distance = $CP + r = 5 + 1 = 6$

- **Length of intercept of the circle on co-ordinate axis :** The intercept made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on x-axis is $AB = 2\sqrt{g^2 - c}$



- The intercept made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on y-axis $\overline{CD} = 2\sqrt{f^2 - c}$



- If the circle $S = 0$ touching x-axis then $g^2 = c$ and equation of the circle is of the form is $x^2 + y^2 + 2gx + 2fy + g^2 = 0$
- If the circle $S = 0$ touching y-axis then $f^2 = c$ and equation of the circle is of the form is $x^2 + y^2 + 2gx + 2fy + f^2 = 0$
- If the circle $S = 0$ touching both the coordinate axes then $g^2 = f^2 = c$ and equation of the circle of the form $x^2 + y^2 \pm 2\sqrt{c}x \pm 2\sqrt{c}y + c = 0$

W.E-10: The equation of the circles which pass through the origin and makes intercepts of lengths 4 and 8 on the x and y axis respectively are

Sol: Circles passes through origin then $c = 0$

$2\sqrt{g^2 - c} = 4$

$2\sqrt{f^2 - c} = 8$

$\Rightarrow \sqrt{g^2 - c} = 2$

$\Rightarrow \sqrt{f^2 - c} = 4$

$\Rightarrow g^2 - 0 = 4$

$\Rightarrow f^2 - 0 = 16$

$\Rightarrow g = \pm 2$

$\Rightarrow f = \pm 4$

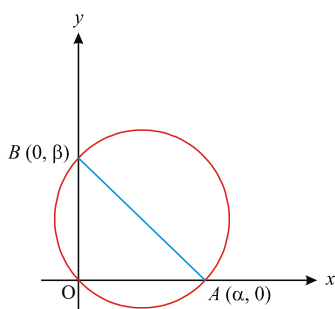
The equation of required circle is

$x^2 + y^2 \pm 4x \pm 8y = 0$

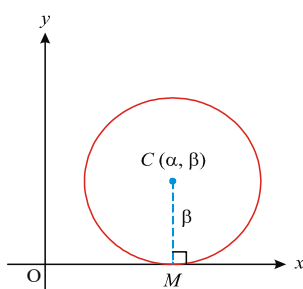
- **Different forms of the equation of circle**
- i) When the circle passes through the origin $(0,0)$ and has intercepts α and β on the x-axis and y-axis, respectively:

$$(x - \alpha)(x - 0) + (y - 0)(y - \beta) = 0$$

i.e., $x^2 + y^2 - \alpha x - \beta y = 0$

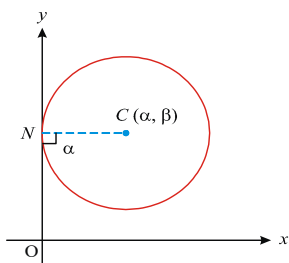


ii) When the circle touches x - axis:



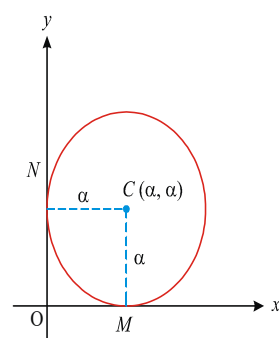
$$(x - \alpha)^2 + (y - \beta)^2 = \beta^2$$

iii) When the circle touches y - axis



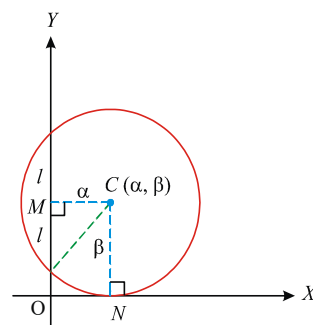
$$(x - \alpha)^2 + (y - \beta)^2 = \alpha^2$$

iv) When the circle touches both axes:



$$(x - \alpha)^2 + (y - \alpha)^2 = \alpha^2$$

v) When the circle touches x - axis at $(\alpha, 0)$ and cuts off intercept on y - axis of length $2l$

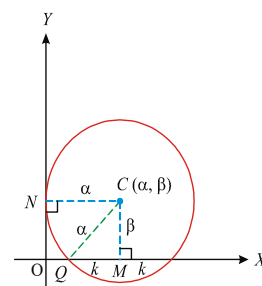


From the figure, $\beta = \sqrt{\alpha^2 + l^2}$

Hence, equation of circle is

$$(x - \alpha)^2 + (y - \beta)^2 = \beta^2$$

vi) When the circle touches y - axis at $(0, \beta)$ and cuts off intercept on x-axis of length $2k$:

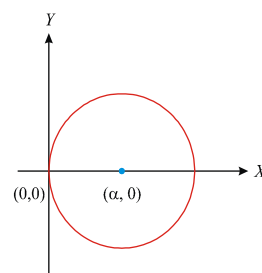


From the figure, $\alpha = \sqrt{\beta^2 + k^2}$

Hence, equation of circle is

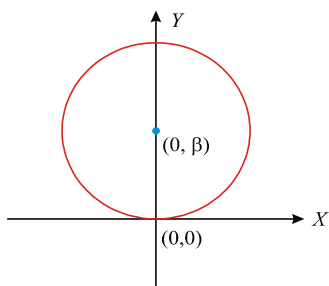
$$(x - \alpha)^2 + (y - \beta)^2 = \alpha^2$$

vii) When the circle passes through origin $(0, 0)$ and centre lies on x-axis :



i.e., $(x - \alpha)^2 + (y - 0)^2 = \alpha^2$

- vii) When the circle passes through origin $(0, 0)$ and centre lies on y-axis :



$$\text{i.e., } (x-0)^2 + (y-\beta)^2 = \beta^2$$

W.E-11: The equation of the circle of radius 3 that lies in 4th quadrant and touching the lines $x=0, y=0$ is

Sol: Circle touches both the coordinate axes and radius

$r=3$ and center is $(3, -3)$.

Equation of the circle is

$$(x-h)^2 + (y-k)^2 = r^2$$

$$\Rightarrow (x-3)^2 + (y+3)^2 = 3^2$$

$$\Rightarrow x^2 + y^2 - 6x + 6y + 9 = 0$$

W.E-12: The equation of the circle which touches both the axes and the straight line $4x+3y=6$ in the first quadrant and lies below it is

Sol: The circle touches both the axes and the straight line $4x+3y=6$ in first quadrant, therefore coordinates of its centre are (a, a) where $a > 0$ since $4x+3y-6=0$ touches the circle,

$$\text{there } \frac{7a-6}{\sqrt{16+9}} = \pm a \Rightarrow 7a-6 = \pm 5a \Rightarrow a = 3, \frac{1}{2}$$

If $a = \frac{1}{2}$ then the equation of required circle is

$$\left(x - \frac{1}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^2 \quad (\text{or})$$

$$4x^2 + 4y^2 - 4x - 4y + 1 = 0$$

➤ **Parametric equations of the circle**

Parametric equations of a curve describes the coordinates of a point on the curve in terms of a single variable. We call this single variable as a parameter.

- i) The parametric equations of a circle with centre (h, k) and radius $r(>0)$ are given by $x = h + r \cos \theta$, $y = k + r \sin \theta$, where $0 \leq \theta < 2\pi$. and the parametric point of this circle is $(h + r \cos \theta, k + r \sin \theta)$.

- ii) The parametric equations of the circle with centre $(0, 0)$ and radius ' r ' is $x = r \cos \theta$, $y = r \sin \theta$ where $0 \leq \theta < 2\pi$. and the parametric point of this circle is $(r \cos \theta, r \sin \theta)$

- Equation of the chord joining the two points θ_1 and θ_2 on the circle $x^2 + y^2 = r^2$ is

$$x \cos \left(\frac{\theta_1 + \theta_2}{2} \right) + y \sin \left(\frac{\theta_1 + \theta_2}{2} \right) = r \cos \left(\frac{\theta_1 - \theta_2}{2} \right)$$

- The length of chord $\overline{AB}$ joining $A(\theta_1), B(\theta_2)$ of the circle $x^2 + y^2 = r^2$ (or)

$$(x-x_1)^2 + (y-y_1)^2 = r^2 \text{ is } 2r \left| \sin \frac{(\theta_1 - \theta_2)}{2} \right|$$

- The parametric angle differ by ' θ ' if $\theta_1 - \theta_2 = \theta$ then AB subtends an angle ' θ ' at center of the circles.

W.E-13: Find the parametric form of the equation of the circle $x^2 + y^2 + px + py = 0$.

Sol: Equation of the circle can be rewritten in the form

$$\left(x + \frac{p}{2}\right)^2 + \left(y + \frac{p}{2}\right)^2 = \frac{p^2}{2}.$$

Therefore, the parametric form of the equation of the given circle is

$$x = \frac{p}{2} + \frac{p}{\sqrt{2}} \cos \theta = \frac{p}{2} (-1 + \sqrt{2} \cos \theta) \quad \text{and}$$

$$y = \frac{p}{2} + \frac{p}{\sqrt{2}} \sin \theta = \frac{p}{2} (-1 + \sqrt{2} \sin \theta) \quad \text{where}$$

$$0 \leq \theta < 2\pi.$$

- **Notation:** $S = x^2 + y^2 + 2gx + 2fy + c$, then

$$S_1 = xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + c$$

$$S_{12} = x_1x_2 + y_1y_2 + g(x_1+x_2) + f(y_1+y_2) + c$$

$$S_{11} = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$$

➤ **Position of a point w.r.t. the circle :**

If $S=0$ is a circle $P(x_1, y_1)$ is a point, then

- i) $S_{11} > 0 \Leftrightarrow P$ lies outside the circle
- ii) $S_{11} = 0 \Leftrightarrow P$ lies on the circle
- iii) $S_{11} < 0 \Leftrightarrow P$ lies inside the circle

W.E-14: Locate the position of the point

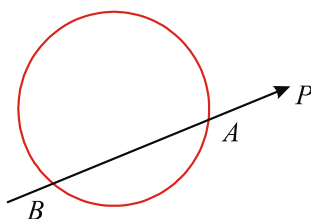
(2, 4) w.r.t the circle $x^2 + y^2 - 4x - 6y + 11 = 0$.

Sol: $S_{11} = 2^2 + 4^2 - 4(2) - 6(4) + 11$

$$= 4 + 16 - 8 - 24 + 11 = -1 < 0 \text{ (-ve)}$$

Since $S_{11} < 0$, the point (2, 4) lies inside the given circle.

- **Power of a point(Def):** If 'C' is centre and 'r' is radius and let P be any point on the plane of a circle then the value of $CP^2 - r^2$ is called the power of the point P w.r.t. circle $S = 0$. (or)
If a secant line through P cuts the circle in any two points A, B then $PA \cdot PB$ is called as power of P .



$$PA \cdot PB = S_{11}$$

- If $S = x^2 + y^2 + 2gx + 2fy + c = 0$, is a circle in standard form and $P(x_1, y_1)$ is a point, then the power of P is S_{11} .

W.E-15: Find the power of the point $P(2, 3)$ w.r.t. the circle $x^2 + y^2 - 2x + 8y - 23 = 0$.

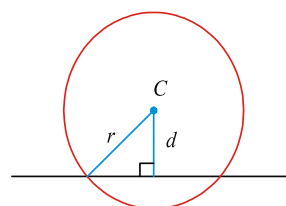
Sol: The power of $P(2, 3)$ w.r.t $S = 0$ is

$$S_{11} = 4 + 9 - 4 + 24 - 23 = 10$$

➤ **Some important points on power:**

- i) power of a point on the circle is zero.
- ii) power is positive if point lies outside the circle.
- iii) power is negative if point lies inside the circle.

- If $P(x_1, y_1)$ is a point lying outside of the circle $S=0$ then the length of the tangent from P is $\sqrt{S_{11}}$.
- The equation of the chord joining the two points (x_1, y_1) and (x_2, y_2) on the circle $s=0$ is $S_1 + S_2 = S_{12}$.
- If 'r' is radius of a circle then a line which is at a distance 'd' from centre of the circle, cuts (a chord) an intercept of length $2\sqrt{r^2 - d^2}$



W.E-16: Find the length of the chord intercepted by the circle $x^2 + y^2 - x + 3y - 22 = 0$ on the line $y = x - 3$.

Sol: Given circle is $x^2 + y^2 - x + 3y - 22 = 0$..(1)

$$\text{Centre of (1), } C = \left(\frac{1}{2}, -\frac{3}{2}\right)$$

$$\text{Radius of (1)} = r = \sqrt{\frac{1}{4} + \frac{9}{4} + 22} = \sqrt{\frac{49}{2}} = \frac{7}{\sqrt{2}}$$

Given line AB is $x - y - 3 = 0$ (2)

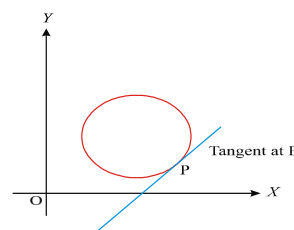
Perpendicular distance from C to (2)

$$CD = d = \left| \frac{\frac{1}{2} + \frac{3}{2} - 3}{\sqrt{2}} \right| = \frac{1}{\sqrt{2}}; \text{ Length of chord,}$$

$$AB = 2\sqrt{r^2 - d^2} = 2\sqrt{\frac{49}{2} - \frac{1}{2}} = 4\sqrt{6}.$$

Tangent Line

- If a line touches the circle at only one point is called a tangent to the circle and the point of coincidence is called the point of contact.



- The equation of the tangent line at (x_1, y_1) on the circle $S = 0$ is $S_1 = 0$.
- The equation of the tangent line at (x_1, y_1) to the circle $x^2 + y^2 = a^2$ is $xx_1 + yy_1 - a^2 = 0$.
- Equation of the tangent at ' θ ' to the circle $x^2 + y^2 = r^2$ is $x \cos \theta + y \sin \theta = r$.
- **Conditions for a line to be a tangent to a circle**
- i) $y = mx + c$ touches $x^2 + y^2 = r^2$ then $c^2 = r^2(m^2 + 1)$
- ii) The equation of tangent to the circle $x^2 + y^2 = r^2$ whose slope ' m ' is $y = mx \pm r\sqrt{m^2 + 1}$
- iii) $lx + my + n = 0$ touches $x^2 + y^2 = r^2$ then $n^2 = r^2(l^2 + m^2)$ and the point of contact is $\left(-\frac{r^2 l}{n}, -\frac{r^2 m}{n}\right)$
- iv) $lx + my + n = 0$ touches $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(l^2 + m^2)(g^2 + f^2 - c) = (lg + mf - n)^2$.
- v) The point of contact of tangent line is foot of perpendicular from centre of the circle.
- vi) For the circle $(x - x_1)^2 + (y - y_1)^2 = r^2$, tangent with slope ' m ' is $(y - y_1) = m(x - x_1) \pm r\sqrt{m^2 + 1}$.
- vii) For the circle $(x - x_1)^2 + (y - y_1)^2 = r^2$ tangents parallel to $lx + my + n = 0$ are $lx + my = lx_1 + my_1 \pm r\sqrt{l^2 + m^2}$
- viii) Let m_1, m_2 are the slopes of tangents through a point $P(x_1, y_1)$ to the circle $x^2 + y^2 = a^2$ then $m_1 + m_2 = \frac{2x_1 y_1}{x_1^2 - a^2}$ and $m_1 m_2 = \frac{y_1^2 - a^2}{x_1^2 - a^2}$.

W.E-17: Find the equation of the circle with centre $(2, 3)$ and touching the line $3x - 4y + 1 = 0$.

Sol. Centre of the circle, $C = (2, 3)$

Given line is $3x - 4y + 1 = 0$ (1)

Since (1) is a tangent to the circle, radius of the circle = perpendicular distance from C to (1)

$$= \frac{|6 - 12 + 1|}{5} = 1$$

Equation of the required circle is

$$(x - 2)^2 + (y - 3)^2 = 1$$

$$\Rightarrow x^2 + y^2 - 4x - 6y + 12 = 0.$$

W.E-18: Find the equations of the tangents to the circle $x^2 + y^2 - 4x - 6y + 3 = 0$ which are inclined at 45° with x-axis.

Sol. Slope of tangent $m = \tan 45^\circ = 1$

Centre of the circle = $(x_1, y_1) = (2, 3)$

Radius of the circle, $r = \sqrt{4 + 9 - 3} = \sqrt{10}$

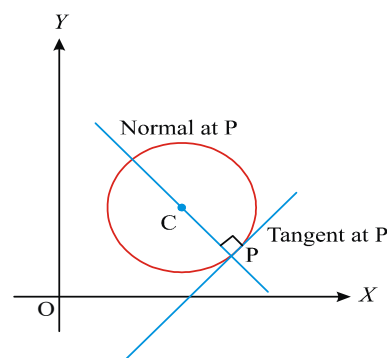
$$y - y_1 = m(x - x_1) \pm r(1 + m^2)$$

$$\Rightarrow y - 3 = 1(x - 2) \pm \sqrt{10}\sqrt{1 + 1}$$

$$\Rightarrow x - y + 1 \pm 2\sqrt{5} = 0.$$

Normal Line

- The normal to a circle is a line which is perpendicular to the tangent at the point of contact.



- The equation of the normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at $P(x_1, y_1)$ is $(x - x_1)(y_1 + f) - (y - y_1)(x_1 + g) = 0$.
- The normal at any point on the circle passing through the centre of the circle.

- The equation of normal to the circle $x^2 + y^2 = r^2$ at $P(x_1, y_1)$ is $xy_1 - yx_1 = 0$.
- Equation of the normal at ' θ ' to the circle $x^2 + y^2 = r^2$ is $x \sin \theta - y \cos \theta = 0$.

W.E-19: Find the equation of the normal to the circle $x^2 + y^2 - 4x - 6y + 11 = 0$ at $(3, 2)$.

Sol. Let $C(2, 3)$ be the centre of given circle. Let $A(3, 2)$, Let the normal at A meets the circle at $B(a, b)$. Since the normal to the circle always passes through C . The equation of the normal at $A(3, 2)$ is $(x-3)(2-3) - (y-2)(3-2) = 0$
i.e., $x + y - 5 = 0$.

W.E-20: If the line $y - x + 3 = 0$ is a normal to the circle $(x-3)^2 + y^2 = 9$ then the point of contact is

Sol. Given $y - x + 3 = 0 \Rightarrow y = x - 3$

substitute $(x-3)^2 + y^2 = 9$.

we get $x = 3 + \frac{3}{\sqrt{2}}$ and $y = \frac{3}{\sqrt{2}}$

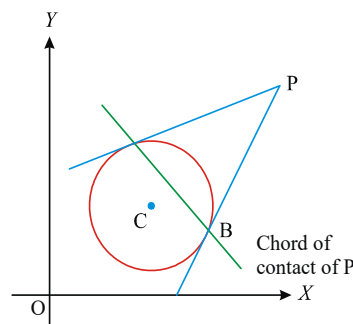
$\therefore$ the point of contact is $\left(3 + \frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$

➤ **Position of a line w.r.t. the circle :**

If ' c ' is centre " r " is radius of a circle and the perpendicular distance from centre C to a line ' l ' is d , then

- $d > r \Leftrightarrow$ ' l ' is outside of the circle.
 - $d = r \Leftrightarrow$ ' l ' is a tangent
 - $d < r \Leftrightarrow$ ' l ' cuts the circle in two distinct points
- The number of circles touching all the three given lines, which are
 - forming a triangle is Four
 - such that two of the lines are parallel is Two
 - all parallel is zero
 - In a circle, normal at any point of the circle passes through its centre.

- **Chord of contact(Def):** The line joining the points of contacts of the tangents to a circle $S = 0$ drawn from an external point P is called chord of contact of P with respect to $S = 0$.



The chord of contact of $P(x_1, y_1)$ with respect to circle $S = 0$ is $S_1 = 0$

- The chord of contact of P becomes tangent at P when P lies on the circle.
- If $P(x_1, y_1)$ lies inside the circle. Then the chord of contact of P does not exist.

W.E-21: Find the equation of the chord of contact of $(0, 5)$ w.r.t. $x^2 + y^2 - 5x + 4y - 2 = 0$.

Sol. Required equation is $S_1 = 0$.

$$\Rightarrow x(0) + y(5) - \frac{5}{2}(x+0) + 2(y+5) - 2 = 0$$

$$\Rightarrow 5x - 14y - 16 = 0$$

W.E-22: The line $9x + y - 28 = 0$ is the chord of contact of the point $p(h, k)$ with respect to the circle $2x^2 + 2y^2 - 3x + 5y - 7 = 0$. Then $p(h, k)$ is

Sol. An equation of the chord of contact of p with respect to the given circle is

$$2xh + 2yk - \frac{3}{2}(x+h) + \frac{5}{2}(y+k) - 7 = 0$$

$$\Rightarrow x\left(2h - \frac{3}{2}\right) + y\left(2k + \frac{5}{2}\right) - \frac{3}{2}h + \frac{5}{2}k - 7 = 0 \dots (1)$$

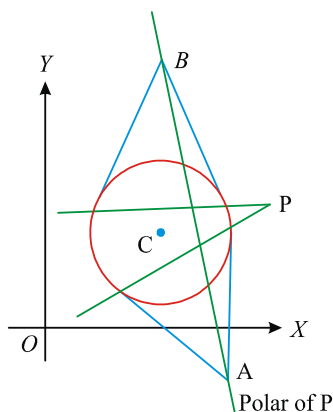
which should be the same as given line $9x + y - 28 = 0 \dots (2)$
Comparing (1) and (2) we get

$$\frac{4h-3}{18} = \frac{4k+5}{2} = \frac{3h-5k+14}{56}$$

So, the point P is $(3, -1)$ but $S = (3, -1) < 0$

where $S = 2x^2 + y^2 - 3x + 5y - 7$ showing that P lies inside the given circle and then cannot be the required point. Thus, the given line cannot be the chord of contact of the given circle for any point P .

- **Pole and polar (Def):** The points of intersection of tangents to the circle $S = 0$ drawn at the extremities of the chords which are passing through a fixed point P lies on the line $L = 0$ then the line $L = 0$ is called the polar of P with respect to $S = 0$. The point P is called pole of the line $L = 0$ with respect to $S = 0$.



- The Polar equation of the point $P(x_1, y_1)$ with respect to the circle $S = 0$ is $S_1 = 0$.
- **Some important aspects of pole and polar**
- If P lies outside the circle, then polar is chord of contact of P .
 - If P lies on the circle, then polar is tangent at P to the circle.
 - If P lies inside the circle, then polar completely lies outside the circle.
 - The polar of centre of circle w.r.t the same circle does not exist.
 - pole of diameter does not exist.
 - polar is perpendicular to the line joining centre and pole.
 - The polar of a point if exists is unique.
 - polars of collinear points are concurrent.
 - poles of concurrent lines are collinear.

- pole of line $lx + my + n = 0$ with respect to the circle $x^2 + y^2 = r^2$ is $\left[-\frac{r^2 l}{n}, -\frac{r^2 m}{n} \right]$.

- The pole of the line $lx + my + n = 0$ w.r.t the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$$\left(-g + \frac{lr^2}{lg + mf - n}, -f + \frac{mr^2}{lg + mf - n} \right)$$

- **Conjugate points (Def):** Two points are said to be conjugate points if polar of one point passes through other point.

$P(x_1, y_1)$ and $Q(x_2, y_2)$ are conjugate points with respect to the curve (circle) $S = 0 \Leftrightarrow S_{12} = 0$.

- **Conjugate lines (Def):** If pole of the line l_1 with respect to a circle lies on the line l_2 , then the pole of l_2 with respect to same circle lies on l_1 . Two such lines are called conjugate lines.

The condition that the lines $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$ to be conjugate w.r.t the circle.

i) $x^2 + y^2 = r^2$ is $r^2(l_1l_2 + m_1m_2) = n_1n_2$

ii) $x^2 + y^2 + 2gx + 2fy + c = 0$ is

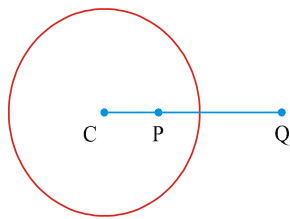
$$(g^2 + f^2 - c)(l_1l_2 + m_1m_2) =$$

$$(l_1g + m_1f - n_1)(l_2g + m_2f - n_2).$$

W.E-23: If the lines $kx + 3y - 1 = 0$, $2x + y + 5 = 0$, are conjugate w.r.t. the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ then find ' k '.

Sol. $r^2(l_1l_2 + m_1m_2) =$
 $= (l_1g + m_1f - n_1)(l_2g + m_2f - n_2)$
 $\Rightarrow 9(2k + 3) = (k + 5)9 \Rightarrow k = 2$

- **Inverse Points(Def):** Let ' C ' be the centre and ' r ' be the radius of the circle $S = 0$. Two points P and Q are said to be inverse points w.r.t. the circle $S = 0$. If
- C, P, Q are collinear
 - P, Q lies on the same side of C
 - $CP \cdot CQ = r^2$



➤ **Some important aspects of inverse points:**

- i) If P lies inside a circle, then its inverse point lies outside the circle
- ii) If P lies on the circle then inverse point of P is itself.
- iii) Inverse point of centre does not exist.
- iv) Inverse points are conjugate points, but the converse need not be true.
- v) Inverse point of a point if exists is unique..
- vi) The inverse point of P with respect to the circle $S = 0$ is the foot of the perpendicular from P or from the centre of the circle to the polar of P.

W.E-24 : If $(6, 8), (k, 2)$ are inverse points w.r.t the circle $x^2 + y^2 = 25$, then $2k =$

Sol. $(x_1, y_1) = (6, 8); \quad (x_2, y_2) = (k, 2)$

The points are inverse w.r.t the circle $x^2 + y^2 = 25$

$$\text{then } S_{12} = 0. \quad \Rightarrow x_1 x_2 + y_1 y_2 - 25 = 0$$

$$\Rightarrow (6)(k) + 8(2) - 25 = 0 \Rightarrow 2k = 3$$

- The inverse point of (x_1, y_1) with respect to the

$$\text{circle } x^2 + y^2 = r^2 \text{ is } \left[\frac{r^2 x_1}{x_1^2 + y_1^2}, \frac{r^2 y_1}{x_1^2 + y_1^2} \right].$$

- The inverse point of origin with respect to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$$\left[-\frac{gc}{g^2 + f^2}, -\frac{fc}{g^2 + f^2} \right]$$

- Equation of the chord of the circle $S=0$ whose mid point is (x_1, y_1) is $S_1 = S_{11}$ and its length is $2\sqrt{|S_{11}|}$.

- Equation of the pair of tangents from (x_1, y_1) to the circle $S = 0$ is $S_1^2 = S_{11}S$.

- Equation of the pair of tangents from origin to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$$(gx + fy)^2 = c(x^2 + y^2).$$

- **Director circle:** Locus of point of intersection of perpendicular tangents to the circle

i) $x^2 + y^2 = r^2$ is $x^2 + y^2 = 2r^2$

ii) $(x - x_1)^2 + (y - y_1)^2 = r^2$ is

$$(x - x_1)^2 + (y - y_1)^2 = 2r^2.$$

- The locus of the point of intersection of perpendicular tangents drawn one to each of the circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ is $x^2 + y^2 = a^2 + b^2$

- The locus of point of intersection of two tangents which include an angle θ , w.r.t the circle $x^2 + y^2 = r^2$ is $x^2 + y^2 = r^2 \sec^2 \theta / 2$.

- The locus of mid points of chords of circle $x^2 + y^2 = r^2$ which subtend angle ' θ ' at centre is $x^2 + y^2 = r^2 \cos^2 \theta / 2$.

- The locus of mid points of chords of the circle $x^2 + y^2 = r^2$ of length '2k' units is $x^2 + y^2 = r^2 - k^2$.

- The locus of the midpoints of the chords of the circle $S=0$ passing through a given point is a circle with the given point and the centre of $S=0$ as ends of diameter.

- The locus of mid points of chords of $x^2 + y^2 = r^2$ which are at a distance of k units from centre is $x^2 + y^2 = k^2$

- If the circle $S=0$ and $L=0$ is a line intersecting the circle in A and B, then any circle passing through A, B is of the form $S + \lambda L = 0$.

- Angle between the tangents from $p(x_1, y_1)$ to

$$S = 0 \text{ is } \theta \text{ then } \tan\left(\frac{\theta}{2}\right) = \frac{r}{\sqrt{S_{11}}}.$$

- If two tangents are perpendicular then $S_{11} = r^2$

- $\overline{PA}, \overline{PB}$ are tangents to circle $S = 0$ with centre C. Then circum centre of $\triangle PAB$ is mid point of CP. Circum radius is $\frac{CP}{2}$.

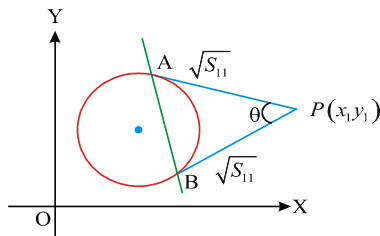
- Area of the quadrilateral formed by the two tangents drawn from an external point to a circle and a pair of radii through their points of contact is $r(\sqrt{S_{11}})$.

- The length of the chord of contact of the point

$$P(x_1, y_1) \text{ w.r.t to the circle } S=0 \text{ is } 2r \sqrt{\frac{S_{11}}{S_{11} + r^2}}.$$

- $S=0$ is a circle in standard form, with centre C and radius r . If $P(x_1, y_1)$ is a point then the area of the triangle formed by pair of tangents from P and chord

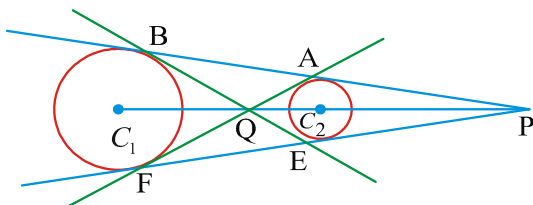
of contact of P is $\frac{r(S_{11})^{3/2}}{S_{11} + r^2}$



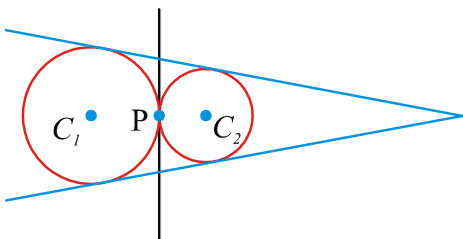
➤ **Relative positions of two circles:**

Let C_1, C_2 be the centres and r_1, r_2 are the radii of the circles $S = 0, S' = 0$ respectively.

- a) If $C_1C_2 > r_1 + r_2$ then the two circles do not intersect. Then we can draw four common tangents to the circles.

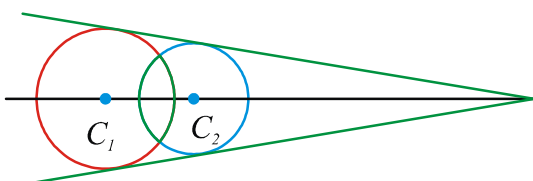


- b) If $C_1C_2 = r_1 + r_2$ then the two circles touch each other externally. Then we can draw three common tangents to the circles

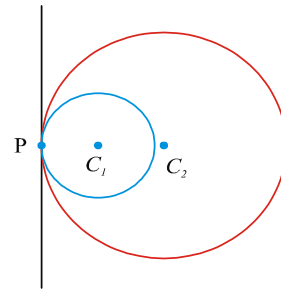


The point of contact P divides C_1C_2 in the ratio $r_1 : r_2$ internally.

- c) If $|r_1 - r_2| < C_1C_2 < r_1 + r_2$ then the two circles intersect each other. Then we can draw two common tangents to the circles.

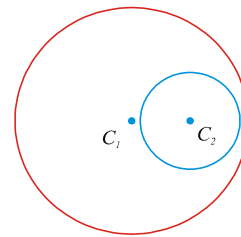


- d) If $C_1C_2 = |r_1 - r_2|$ then the two circles touch each other internally. Then we can draw one common tangent to the circles.



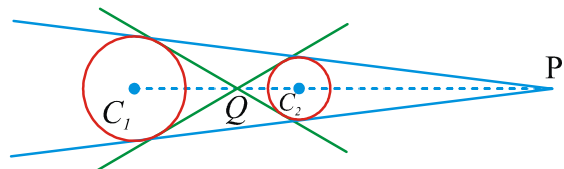
The point of contact P divides $\overline{C_1C_2}$ in the ratio $r_1 : r_2$ externally.

- e) If $C_1C_2 < |r_1 - r_2|$ then one circle entirely lies inside the other circle. Then we cannot draw common tangent.



- Two circles with centres C_1, C_2 and radii r_1, r_2 touch each other if $C_1C_2 = |r_1 \pm r_2|$

- **Centres of similitude:** The point of intersection of direct common tangents of two circles (P) is called as external centre of similitude. The point of intersection of transverse common tangents of two circles (Q) is called as internal centre of similitude.



The internal center of similitude (Q) divides C_1C_2 in the ratio $r_1 : r_2$ internally and the external centre of similitude (P) divides C_1C_2 in the ratio $r_1 : r_2$ externally.

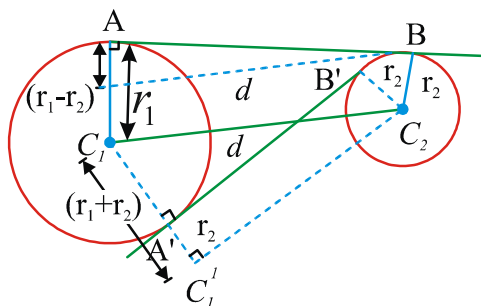
- The circle with centres of similitudes as diameter is called circle of similitude.

➤ **Length of common tangents :**

If d is the distance between centres of two circles whose radii are r_1 and r_2 then length of direct common tangent of two circles is

$AB = \sqrt{d^2 - (r_1 - r_2)^2}$ and length of transverse common tangent of two circles is

$$A'B' = \sqrt{d^2 - (r_1 + r_2)^2}.$$

**W.E.-25:** From the point $A(0, 3)$ on the circle

$x^2 + 4x + (y - 3)^2 = 0$, a chord AB drawn and extended to a point M such that $AM = 2AB$. Find the equation of the locus of M .

Sol: Let $M = (x_1, y_1)$

Given that $AM = 2AB \Rightarrow AB = BM$

$$\Rightarrow B \text{ is the midpoint of } AM \quad B = \left(\frac{x_1}{2}, \frac{y_1 + 3}{2} \right)$$

B is a point on the given circle

$$\left(\frac{x_1}{2} \right)^2 + 4 \left(\frac{x_1}{2} \right) + \left(\frac{y_1 + 3}{2} - 3 \right)^2 = 0$$

$$\Rightarrow \frac{x_1^2}{4} + 2x_1 + \frac{y_1^2 - 6y_1 + 9}{4} = 0$$

$$\Rightarrow x_1^2 + y_1^2 + 8x_1 - 6y_1 + 9 = 0$$

Locus of M is $x^2 + y^2 + 8x - 6y + 9 = 0$ which is a circle

W.E.-26: A square is inscribed in the circle

$x^2 + y^2 + 2x + 4y + 3 = 0$. Its sides are parallel to the coordinate axes, then find the vertices of a square.

Sol: Centre $(-1, -2)$, radius $r = \sqrt{2}$.

$$\text{Vertices of a square} = \left(-1 \pm \sqrt{2} \cdot \frac{1}{\sqrt{2}}, -2 \pm \sqrt{2} \cdot \frac{1}{\sqrt{2}} \right)$$

$$= (-1 \pm 1, -2 \pm 1) = (0, -1)(0, -3)(-2, -1)(-2, -3)$$

W.E.-27: Show that the equation of the circle with centre at origin and passing through the vertices of an equilateral triangle whose median is of length $3a$ is $x^2 + y^2 = 4a^2$.

Sol: Clearly centre of a circle is $(0, 0)$

$$\text{Radius of the circle is } \frac{2}{3}(3a) = 2a$$

$$\text{Equation of the circle is } x^2 + y^2 = 4a^2.$$

W.E.-28: If $\left(m_i, \frac{1}{m_i} \right), m_i > 0, i = 1, 2, 3, 4$ are four distinct points on a circle, show that $m_1 m_2 m_3 m_4 = 1$.

Sol: Let equation of the circle

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{Circle passes through } \left(m_i, \frac{1}{m_i} \right)$$

$$\Rightarrow m_i^4 + 2gm_i^3 + cm_i^2 + 2fm_i + 1 = 0; m_1 m_2 m_3 m_4 = 1$$

W.E.-29 : A circle is inscribed in an equilateral triangle of side a . Show that the area of the square inscribed in the circle is $\frac{a^2}{6}$.

Sol: Let radius of the circle is r then length of the altitude

$$\text{of a triangle} = 3r = \frac{\sqrt{3}}{2}a \Rightarrow r = \frac{a}{2\sqrt{3}}$$

$$\text{Diagonal of the square is } \frac{a}{\sqrt{3}}$$

$$\text{Side of the square is } \frac{a}{\sqrt{6}}.$$

$$\text{Area of the square is } \frac{a^2}{6}.$$

W.E.-30: Each side of $\triangle ABC$ is the polar of the opposite vertex with respect to a circle with centre P . Then P is ortho centre of $\triangle ABC$.

Sol: Since line joining vertex A and centre P is perpendicular polar of $P(BC)$.

P is orthocentre of $\triangle ABC$.

In this case A, B, C are pairwise conjugate points w.r.to the circle



C.U.Q

- Number of circles drawn through two points is
 - One
 - Two
 - Three
 - Infinite
- If three lines are not concurrent and no two of them are parallel, number of circles drawn touching all the three lines
 - 1
 - 4
 - 3
 - Infinite
- Equation of circle passing through non-collinear points A, B, C is
 - Equation of the circle on AB as diameter + K (equation of AB) = 0
 - Equation of the circle on AB as diameter + K (equation of BC) = 0
 - Equation of the circle on AB as diameter + K (equation of CA) = 0
 - Equation of the circle on BC as diameter + K (equation of AC) = 0
- The equation of the chord joining α, β on the circle $S = 0$ is
 - $(x+g)\cos\left(\frac{\alpha+\beta}{2}\right) + (y+f)\sin\left(\frac{\alpha+\beta}{2}\right) = \cos\left(\frac{\alpha-\beta}{2}\right)$
 - $(x+g)\cos\left(\frac{\alpha+\beta}{2}\right) + (y+f)\sin\left(\frac{\alpha+\beta}{2}\right) = r\cos\left(\frac{\alpha-\beta}{2}\right)$
 - $(x-g)\cos\left(\frac{\alpha+\beta}{2}\right) + (y-f)\sin\left(\frac{\alpha+\beta}{2}\right) = r\cos\left(\frac{\alpha-\beta}{2}\right)$
 - $(x+g)\cos\left(\frac{\alpha-\beta}{2}\right) + (y+f)\sin\left(\frac{\alpha-\beta}{2}\right) = r\cos\left(\frac{\alpha-\beta}{2}\right)$
- If the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ touches x-axis at $(x_1, 0)$ then $x^2 + 2gx + c =$
 - $(x - x_1)^2$
 - $(x + x_1)^2$
 - $(y - y_1)^2$
 - $(y + y_1)^2$
- G.M of shortest and farthest distances from a point w.r.t a circle $S = 0$ is
 - S_{11}
 - $\sqrt{|S_{11}|}$
 - S_{11}^2
 - $\sqrt{2r}$
- The locus of the point of intersection of the two tangents drawn to the circle $x^2 + y^2 = a^2$ which include an angle α is
 - $x^2 + y^2 = a^2 \operatorname{cosec}^2 \alpha / 2$
 - $x^2 + y^2 = a^2 \cot^2 \alpha / 2$
 - $x^2 + y^2 = a^2 \tan \alpha$
 - $x^2 + y^2 = a^2 \tan \alpha / 2$
- A and B are two fixed points. The locus of P such that in $\triangle PAB$, $\frac{\sin B}{\sin A}$ is a constant ($\neq 1$) is _____
 - a circle
 - pair of lines
 - part of a circle
 - line parallel to BC
- A tangent at a point on the circle $x^2 + y^2 = a^2$ intersect a concentric circle 'S' at P and Q. The tangents of this circle at P, Q meet on the circle $x^2 + y^2 = b^2$ then the equation of concentric circle is
 - $x^2 + y^2 = a^2 b^2$
 - $x^2 + y^2 = ab$
 - $x^2 + y^2 = a^2 + b^2$
 - $x^2 + y^2 = \frac{a}{b}$
- If t_1 and t_2 are the lengths of tangents drawn from two conjugate points A, B then $t_1^2 + t_2^2 =$
 - AB
 - 2AB
 - AB^2
 - $2AB^2$
- The locus of middle points of the chords of the circle $x^2 + y^2 = a^2$ subtending an angle ' α ' at the centre is
 - $x^2 + y^2 = a \cos \alpha / 2$
 - $x^2 + y^2 = a^2 \cos \alpha / 2$
 - $x^2 + y^2 = a^2 \cos^2 \alpha / 2$
 - $x^2 + y^2 = \cos^2 \alpha / 2$
- Pole of diameter of a circle w.r. to the same circle lies
 - inside of the circle
 - outside of the circle
 - on the circle
 - Does not exist
- The equation of tangent at θ on $S = 0$
 - $(x+g)\cos\theta + (y+f)\sin\theta = r$
 - $(x+g)\cos\theta + (y+f)\sin\theta = r^2$
 - $(x-g)\cos\theta + (y-f)\sin\theta = r$
 - $(x-g)\cos\theta + (y-f)\sin\theta = r^2$
- Equation of the circle passing through (0,0), (a, b) and (b, a) is
 - $(a+b)(x^2 + y^2) - (a^2 + b^2)(x+y) = 0$
 - $(a^2 + b^2)(x^2 + y^2) - (a+b)(x+y) = 0$
 - $(a+b)(x^2 + y^2) + (a^2 + b^2)(x+y) = 0$
 - $(a^2 + b^2)(x^2 + y^2) + (a+b)(x+y) = 0$

15. An equilateral triangle is inscribed in the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ then its side is

1) $3r$ 2) $\sqrt{3}r$ 3) $\frac{r}{\sqrt{3}}$ 4) $2r$

16. The polars of three points w.r. to a given circle are concurrent then the three points are

- 1) collinear 2) form an equilateral
3) form a right angled triangle
4) passes through the centre of the circle

17. If $x^2 + y^2 - 2x + 4y + 1 = 0$ complete one revolution on the x-axis then the centre in new position is

- 1) $(1, -2)$ 2) $(1, 4\pi - 2)$
3) $(1 + 4\pi, -2)$ 4) $(1 - 4\pi, -2)$

18. $S = 0$ is a circle and $P(x_1, y_1)$ is an external point to it $\overline{PA}$ and $\overline{PB}$ are tangents to $S = 0$ from the point P, A and B are points of contacts of tangents. The centre of circum circle of ΔPAB is

- 1) $(x_1 - g, y_1 - f)$ 2) $(x_1 + g, y_1 + f)$
3) $\left(\frac{x_1 - g}{2}, \frac{y_1 - f}{2}\right)$ 4) $\left(\frac{-g}{2}, \frac{-f}{2}\right)$

19. A circle is inscribed in an equilateral triangle of side 'a' the area of any square inscribed in this circle is

1) $\frac{a^2}{3}$ 2) $\frac{a^2}{6}$ 3) $\frac{2a^2}{3}$ 4) $\frac{a^2}{\sqrt{3}}$

20. Chords of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ subtends a right angle at the origin. The locus of the feet of the perpendiculars from the origin to these chords is

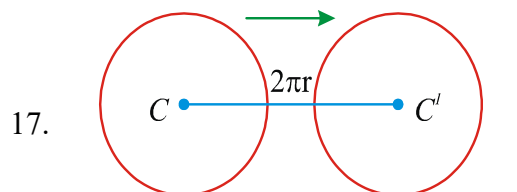
- 1) $x^2 + y^2 + gx + fy + c = 0$
2) $2(x^2 + y^2) + gx + fy + c = 0$
3) $2(x^2 + y^2 + gx + fy) + c = 0$
4) $x^2 + y^2 + 2(gx + fy + c) = 0$

C.U.Q KEY

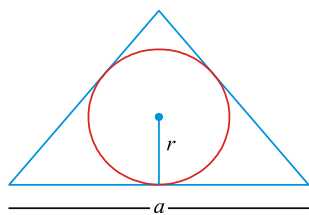
- 1) 4 2) 2 3) 1 4) 2 5) 1
6) 2 7) 1 8) 1 9) 2 10) 3
11) 3 12) 4 13) 1 14) 1 15) 2
16) 1 17) 3 18) 3 19) 2 20) 3

C.U.Q - HINTS

1. Draw a diagram
2. Draw a diagram
3. The equation of the circle on AB as diameter + K(equation of AB)=0
4. The equation of line joining
 $A(-g + r \cos \alpha, -f + r \sin \alpha)$
 $B(-g + r \cos \beta, -f + r \sin \beta)$
5. $x^2 + 2gx + c = 0$ has equal roots are x_1, x_2
6. $\sqrt{(CP + r)(CP - r)} = \sqrt{S_{11}}$
7. $\tan \frac{\alpha}{2} = \frac{a}{\sqrt{S_{11}}}$
8. $\frac{PA}{PB} = \frac{\sin B}{\sin A} = k$
9. take $p(x_1, y_1)$ be any point on $x^2 + y^2 = a^2$ and $Q(\alpha, \beta)$ be on $x^2 + y^2 = b^2$ and the polar of (α, β) w.r.t $x^2 + y^2 = r^2$ touches $x^2 + y^2 = a^2$
10. $t_1^2 + t_2^2 = S_{11} + S_{22} = AB^2$
11. Let $P(x_1, y_1)$ be mid point of chord and
 $\cos \frac{\alpha}{2} = \frac{CP}{a}$
12. The polar of any point on the diameter w.r. to same circle does not exist
13. Find equation of tangent at $(-g + r \cos \theta, -f + r \sin \theta)$
14. Find equation of circle passing through given three points
15. $\cos 30^\circ = \frac{a}{2r}$, a is a side of Δ^e
16. Statement



- 17.
18. Mid point of $\overline{CP}$
19. $r = \frac{h}{3} = \frac{a}{2\sqrt{3}}$ h is length of median $\therefore R = \frac{a}{\sqrt{3}}$



$$\text{Area of square} = \frac{R^2}{2}$$

20. Let $p(x_1, y_1)$ be the mid point of a chord. OP is Perpendicular to the chord. Equation of the chord is $xx_1 + yy_1 = x_1^2 + y_1^2 \rightarrow (1)$
Homogeneous the circle equation with the help of (1). Then use sum of the coefficients of x^2 and y^2 is equal to zero



LEVEL-I (C.W)

EQUATION OF CIRCLE, CENTRE -RADIUS FORM

- The equation of the circle with radius 3 and centre as the point of intersection of the lines $2x + 3y = 5, 2x - y = 1$ is
 - $x^2 + y^2 = 9$
 - $x^2 + y^2 - 2x - 2y - 7 = 0$
 - $x^2 + y^2 - 2x - 2y + 7 = 0$
 - $x^2 + y^2 + 9 = 0$
- The circle concentric with $x^2 + y^2 + 4x + 6y + 3 = 0$ and radius 2 is
 - $x^2 + y^2 + 4x + 6y - 9 = 0$
 - $x^2 + y^2 + 4x + 6y + 9 = 0$
 - $x^2 + y^2 - 4x - 6y + 9 = 0$
 - $x^2 + y^2 = 4$
- Find the equation of the circle passing through $(-2, 14)$ and concentric with the circle $x^2 + y^2 - 6x - 4y - 12 = 0$.
 - $x^2 + y^2 - 6x - 4y - 156 = 0$
 - $x^2 + y^2 - 6x + 4y - 156 = 0$
 - $x^2 + y^2 - 6x + 4y + 156 = 0$
 - $x^2 + y^2 + 6x + 4y + 156 = 0$
- If the centroid of an equilateral triangle is $(1, 1)$ and one of its vertices is $(-1, 2)$ then, equation of its circum circle is
 - $x^2 + y^2 - 2x - 2y - 3 = 0$
 - $x^2 + y^2 + 2x - 2y - 3 = 0$
 - $x^2 + y^2 - 4x - 6y + 9 = 0$
 - $x^2 + y^2 + x - y + 5 = 0$
- For the circle $ax^2 + y^2 + bx + dy + 2 = 0$ center is $(1, 2)$ then $2b + 3d =$
 - 16
 - 16
 - 8
 - 8
- The radius of the circle passing through $(6, 2)$ and the equations of two normals for the circle are $x + y = 6$ and $x + 2y = 4$ is
 - $\sqrt{5}$
 - $2\sqrt{5}$
 - $3\sqrt{5}$
 - $4\sqrt{5}$
- Origin is the centre of circle passing through the vertices of an equilateral triangle whose median is of length $3a$ then equation of the circle is
 - $x^2 + y^2 = a^2$
 - $x^2 + y^2 = 2a^2$
 - $x^2 + y^2 = 3a^2$
 - $x^2 + y^2 = 4a^2$
- The diameters of a circle are along $2x + y - 7 = 0$ and $x + 3y - 11 = 0$. Then the equation of this circle which also passes through $(5, 7)$ is (EAMCET-2009)
 - $x^2 + y^2 - 4x - 6y - 16 = 0$
 - $x^2 + y^2 - 4x - 6y - 20 = 0$
 - $x^2 + y^2 - 4x - 6y - 12 = 0$
 - $x^2 + y^2 + 4x + 6y - 12 = 0$
- If the two circles $x^2 + y^2 + 2gx + c = 0$ and $x^2 + y^2 - 2fy - c = 0$ have equal radius then locus of (g, f) is
 - $x^2 + y^2 = c^2$
 - $x^2 - y^2 = 2c$
 - $x - y^2 = c^2$
 - $x^2 + y^2 = 2c^2$
- Centre and radius of the circle with segment of the line $x + y = 1$ cut off by coordinate axes as diameter is
 - $\left(\frac{1}{2}, \frac{1}{2}\right), \frac{1}{\sqrt{2}}$
 - $\left(-\frac{1}{2}, -\frac{1}{2}\right), \frac{1}{\sqrt{2}}$
 - $\left(\frac{1}{2}, -\frac{1}{2}\right), \frac{1}{\sqrt{2}}$
 - $\left(-\frac{1}{2}, \frac{1}{2}\right), \frac{1}{\sqrt{2}}$

SHORTEST DISTANCE AND LONGEST DISTANCE

11. The shortest distance from $(-2, 14)$ to the circle $x^2 + y^2 - 6x - 4y - 12 = 0$ is
 1) 4 2) 6 3) 8 4) 10
12. The least distance of the line $8x - 4y + 73 = 0$ from the circle $16x^2 + 16y^2 + 48x - 8y - 43 = 0$ is
 1) $\sqrt{5}/2$ 2) $2\sqrt{5}$ 3) $3\sqrt{5}$ 4) $4\sqrt{5}$

CONCYCLIC POINTS, CIRCUMSCRIBING CIRCLES

13. If the points $(0,0)$, $(2,0)$, $(0,-2)$, and $(k,-2)$ are concyclic then $k =$
 1) 2 2) -2 3) 0 4) 1
14. If a circle is inscribed in a square of side 10, so that the circle touches the four sides of the square internally then radius of the circle is
 1) 10 2) $5\sqrt{2}$ 3) $10\sqrt{2}$ 4) 5
15. The centre of the circle circumscribing the square whose three sides are $3x + y = 22$, $x - 3y = 14$ and $3x + y = 62$ is:
 1) $\left(\frac{3}{2}, \frac{27}{2}\right)$ 2) $\left(\frac{27}{2}, \frac{3}{2}\right)$
 3) $(27, 3)$ 4) $\left(1, \frac{2}{3}\right)$
16. If the points $(2,0)$, $(0,1)$, $(4,0)$ and $(0,a)$ are concyclic then $a =$
 1) 2 2) 4 3) 6 4) 8
17. Centre of the circle inscribed in a rectangle formed by the lines $x^2 - 8x + 12 = 0$ and $y^2 - 14y + 40 = 0$ is
 1) $(4, 7)$ 2) $(7, 4)$ 3) $(9, 4)$ 4) $(4, 9)$

POWER OF A POINT, POSITION OF A POINT, CHORD

18. The power of $(1,1)$ with respect to the circle $x^2 + y^2 - 4x + 3y + k = 0$ is 3 then k is
 1) 1 2) 2
 3) 3 4) 4

19. The equation of the circle with centre $(3,2)$ and the power of $(1,-2)$ w.r.t the circle $x^2 + y^2 = 1$ as radius is
 1) $x^2 + y^2 - 6x - 4y - 3 = 0$
 2) $x^2 + y^2 - 3x - 2y - 3 = 0$
 3) $x^2 + y^2 + 6x + 4y - 3 = 0$
 4) $x^2 + y^2 - 6x - 4y + 3 = 0$
20. If a line is drawn through a point $A(3,4)$ to cut the circle $x^2 + y^2 = 4$ at P and Q then $AP \cdot AQ =$
 1) 15 2) 17 3) 21 4) 25
21. A chord of length 24 units is at a distance of 5 units from the center of a circle then its radius is
 1) 5 2) 12 3) 13 4) 10
22. If the line $3x - 4y = \lambda$ cuts the circle $x^2 + y^2 - 4x - 8y - 5 = 0$ in two points then limits of λ are (AIEEE-2010)
 1) $[-35, 15]$ 2) $(-35, 15)$
 3) $(-35, 10)$ 4) $(-35, 15]$

TANGENT AND NORMAL

23. The length of the tangent from $(1,1)$ to the circle $2x^2 + 2y^2 + 5x + 3y + 1 = 0$ is
 1) $\sqrt{13/2}$ 2) 3 3) 2 4) 1
24. The locus of the point from which the length of the tangent to the circle $x^2 + y^2 - 2x - 4y + 4 = 0$ is 3 units is
 1) $x^2 + y^2 - 2x - 4y - 9 = 0$
 2) $x^2 + y^2 - 2x - 4y - 4 = 0$
 3) $x^2 + y^2 - 2x - 4y - 3 = 0$
 4) $x^2 + y^2 - 2x - 4y - 5 = 0$
25. The tangent to the circle $x^2 + y^2 - 4x + 2y + k = 0$ at $(1,1)$ is $x - 2y + 1 = 0$ then $k =$
 1) -1 2) 0 3) 1 4) 2
26. The equations of the tangents to the circle $x^2 + y^2 = 25$ with slope 2 is
 1) $y = 2x \pm \sqrt{5}$ 2) $y = 2x \pm 2\sqrt{5}$
 3) $y = 2x \pm 3\sqrt{5}$ 4) $y = 2x \pm 5\sqrt{5}$

27. The line $4y - 3x + \lambda = 0$ touches the circle $x^2 + y^2 - 4x - 8y - 5 = 0$ then $\lambda =$
 1) 29 2) 10 3) -35 4) 35
28. The circle to which two tangents can be drawn from origin is
 1) $x^2 + y^2 - 8x - 4y - 3 = 0$
 2) $x^2 + y^2 + 4x + 2y + 2 = 0$
 3) $x^2 + y^2 - 8x + 6y + 1 = 0$
 4) both (2) & (3)
29. The normal at (1,1) to the circle $x^2 + y^2 - 4x + 6y - 4 = 0$ is
 1) $4x + 3y = 7$ 2) $4x + y = 5$
 3) $x + y = 2$ 4) $4x - y = 5$
30. Slopes of tangents through (7, 1) to the circle $x^2 + y^2 = 25$ satisfy the equation
 1) $12m^2 + 7m + 12 = 0$
 2) $12m^2 - 7m + 12 = 0$
 3) $12m^2 + 7m - 12 = 0$
 4) $12m^2 - 7m - 12 = 0$
31. Angle between tangents drawn from a point P to circle $x^2 + y^2 - 4x - 8y + 8 = 0$ is 60° then length of chord of contact of P is
 1) 6 2) 4 3) 2 4) 3
32. Locus of the point of intersection of perpendicular tangents to the circle $x^2 + y^2 = 10$ is
 1) $x^2 + y^2 = 5$ 2) $x^2 + y^2 = 20$
 3) $x^2 + y^2 = 10$ 4) $x^2 + y^2 = 100$
33. Locus of the point of intersection of perpendicular tangents drawn one to each of the circles $x^2 + y^2 = 8$ and $x^2 + y^2 = 12$ is
 1) $x^2 + y^2 = 4$ 2) $x^2 + y^2 = 20$
 3) $x^2 + y^2 = 208$ 4) $x^2 + y^2 = 16$
34. The locus of the point of intersection of perpendicular tangents drawn to each of circles $x^2 + y^2 = 16$ and $x^2 + y^2 = 9$ is a circle whose diameter is
 1) 5 2) $\sqrt{7}$
 3) $2\sqrt{7}$ 4) 10

35. The equation of the circle with centre at (4,3) and touching the line $5x - 12y - 10 = 0$ is
 1) $x^2 + y^2 - 4x - 6y + 4 = 0$
 2) $x^2 + y^2 + 6x - 8y + 16 = 0$
 3) $x^2 + y^2 - 8x - 6y + 21 = 0$
 4) $x^2 + y^2 - 24x - 10y + 144 = 0$
36. The locus of points from which lengths of tangents to the two circles $x^2 + y^2 + 4x + 3 = 0$, $x^2 + y^2 - 6x + 5 = 0$ are in the ratio 2 : 3 is a circle with centre
 1) (6,0) 2) (-6,0) 3) (0,6) 4) (0,-6)

CIRCLES TOUCHING COORDINATE AXES AND INTERCEPTS ON AXES

37. The circle with centre (4,-1) and touching x-axis is
 1) $x^2 + y^2 - 8x + 2y + 16 = 0$
 2) $x + y^2 + 18x - 2y - 16 = 0$
 3) $x^2 + y^2 - 4x + y + 4 = 0$
 4) $x^2 + y^2 + 14x - y + 4 = 0$
38. If the line $hx + ky = 1/a$ touches the circle $x^2 + y^2 = a^2$ then the locus of (h,k) is circle of radius
 1) $\frac{1}{a}$ 2) a^2 3) a 4) $\frac{1}{a^2}$
39. The circle $x^2 + y^2 - 2ax - 2ay + a^2 = 0$ touches axes of co-ordinates at
 1) (a,a), (0,0) 2) (a,0), (0,0)
 3) (a,0), (0,a) 4) (0,a), (1,a)
40. The y-intercept of the circle $x^2 + y^2 + 4x + 8y - 5 = 0$ is
 1) $2\sqrt{21}$ 2) $2\sqrt{19}$ 3) 6 4) 12
41. The intercept made by the circle with centre (2, 3) and radius 6 on y-axis is
 1) $18\sqrt{2}$ 2) $12\sqrt{2}$ 3) $8\sqrt{2}$ 4) $6\sqrt{2}$
42. The centre of the circle passing through origin and making intercepts 8 and -4 on x and y-axes respectively is
 1) (4, -2) 2) (-2, 4) 3) (8, -4) 4) both (1) & (2)

43. $2x + y = 0$ is the equation of a diameter of the circle which touches the lines $4x - 3y + 10 = 0$ and $4x - 3y - 30 = 0$. The centre and radius of the circle are

- 1) $(-2, 1); 4$ 2) $(1, -2); 8$
 3) $(1, -2); 4$ 4) $(1, -2); 16$

44. The equation of the circle which has both the axes as its tangents and which passes through the point $(1, 2)$

- 1) $x^2 + y^2 - 2x + 2y - 1 = 0$
 2) $x^2 + y^2 - 2x + 2y + 1 = 0$
 3) $x^2 + y^2 - 2x - 2y + 1 = 0$
 4) $x^2 + y^2 - 2x - 2y - 1 = 0$

45. If a circle of radius 2 touches X-axis at $(1, 0)$ then its centre may be

- 1) $(1, 2)$ and $(1, -2)$ 2) $(1, 2)$ and $(2, 1)$
 3) $(-1, 2)$ and $(1, -2)$ 4) $(-1, 2)$ and $(-1, -2)$

46. Equation of circles touching x-axis at the origin and the line $4x - 3y + 24 = 0$ are

- 1) $x^2 + y^2 - 6y = 0, x^2 + y^2 + 24y = 0,$
 2) $x^2 + y^2 + 2y = 0, x^2 + y^2 - 18y = 0,$
 3) $x^2 + y^2 + 18x = 0, x^2 + y^2 - 8x = 0,$
 4) $x^2 + y^2 + 4x = 0, x^2 + y^2 - 16x = 0$

47. Centre of the circle touching y-axis at $(0, 3)$ and making an intercept 2 units on positive X-axis is

- 1) $(\sqrt{10}, 3)$ 2) $(3, \sqrt{10})$
 3) $(-\sqrt{10}, 3)$ 4) $(-\sqrt{10}, -3)$

PARAMETRIC EQUATIONS

48. Parametric equation of the circle $x^2 + y^2 = 16$ are

- 1) $x = 4 \cos \theta, y = 4 \sin \theta$
 2) $x = 4 \cos \theta, y = 4 \tan \theta$
 3) $x = 4 \cosh \theta, y = 4 \sinh \theta$
 4) $x = 4 \sec \theta, y = 4 \tan \theta$

49. Locus of the point $(\sec h\theta, \tanh \theta)$ is

- 1) $x^2 + y^2 = 1$ 2) $x^2 - y^2 = 1$
 3) $x^2 + y^2 + 1 = 0$ 4) $x^2 - y^2 = x + y$

50. To the circle $x^2 + y^2 + 8x - 4y + 4 = 0$ tangent at the point $\theta = \frac{\pi}{4}$ is

- 1) $x + y + 2 - 4\sqrt{2} = 0$ 2) $x - y + 2 - 4\sqrt{2} = 0$
 3) $x + y + 2 + 4\sqrt{2} = 0$ 4) $x - y - 2 - 4\sqrt{2} = 0$

CHORD OF CONTACT, POLE, POLAR, CONJUGATE POINTS AND LINES, INVERSE POINTS (POLE AND POLAR NOT FOR MAINS)

51. The chord of contact of $(2, 1)$ w.r.t to the circle $x^2 + y^2 + 4x + 4y + 1 = 0$ is

- 1) $2x + y + 7 = 0$ 2) $4x + 3y + 7 = 0$
 3) $3x + 4y + 1 = 0$ 4) not existing

52. The polar of $(2, -1)$ w.r.t $x^2 + y^2 + 6x + 4y - 1 = 0$ is $5x + y + k = 0$ then $k =$

- 1) 1 2) 2 3) 3 4) 4

53. pole of $3x + 5y + 17 = 0$ w.r.t the circle

$x^2 + y^2 + 4x + 6y + 9 = 0$ is

- 1) $(-1, 2)$ 2) $(1, 2)$ 3) $(1, -2)$ 4) $(2, 1)$

54. If $ax + by + c = 0$ is the polar of $(1, 1)$ w.r.t the circle $x^2 + y^2 - 2x + 2y + 1 = 0$ and H.C.F of a, b, c is equal to one then $a^2 + b^2 + c^2 =$

- 1) 0 2) 3 3) 5 4) 15

55. If $(1, 1), (k, 2)$ are conjugate points with respect to the circle $x^2 + y^2 + 8x + 2y + 3 = 0$, then $k =$

- 1) -12 2) -12/7 3) -12/5 4) -4

56. If $3x + 2y = 3$ and $2x + 5y = 1$ are conjugate lines w.r.t the circle $x^2 + y^2 = r^2$ then $r^2 =$

- 1) $\frac{3}{16}$ 2) $\frac{16}{3}$ 3) $\frac{4}{\sqrt{3}}$ 4) $\frac{\sqrt{3}}{4}$

57. The points $(3, 2), (2, 3)$ w.r.t the circle $x^2 + y^2 = 12$ are

- 1) extremities of a diameter 2) conjugate points
 3) Inverse points 4) lie on the circle

58. The length of chord of contact of the point $(3, 6)$ with respect to the circle $x^2 + y^2 = 10$ is

- 1) $\frac{2\sqrt{70}}{3}$ 2) $6\sqrt{5}$ 3) $\sqrt{5}$ 4) $\frac{12}{\sqrt{5}}$

59. If the lines $2x + 3y - 4 = 0$ and $kx + 4y - 2 = 0$ are conjugate with respect to the circle $x^2 + y^2 = 4$ then $k - 1 =$
 1) -5 2) -6 3) -4 4) 5
60. For the circle $x^2 + y^2 - 2x - 4y - 4 = 0$, the lines $2x + 3y - 1 = 0$, $2x + y + 5 = 0$ are
 1) perpendicular tangents
 2) conjugate
 3) parallel tangents
 4) perpendicular chords
61. The inverse point of $(2, -3)$ w.r.t to circle $x^2 + y^2 + 6x - 4y - 12 = 0$ is
 1) $\left(\frac{1}{2}, \frac{1}{2}\right)$ 2) $\left(-\frac{1}{2}, \frac{1}{2}\right)$
 3) $\left(\frac{1}{2}, -\frac{1}{2}\right)$ 4) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$
62. If the inverse of $P(-3, 5)$ w.r.t to a circle is $(1, 3)$ then polar of P w.r.t to the circle is
 1) $x + 2y = 7$ 2) $2x - 2y + 11 = 0$
 3) $2x - y + 1 = 0$ 4) $2x - y - 1 = 0$

CHORD WITH MID POINT

63. The equation of the chord of $x^2 + y^2 - 4x + 6y + 3 = 0$ whose mid point is $(1, -2)$ is
 1) $x + y + 1 = 0$ 2) $2x + 3y + 4 = 0$
 3) $x - y - 3 = 0$ 4) not existing
64. The pair of tangents from $(2, 1)$ to the circle $x^2 + y^2 = 4$ is
 1) $3x^2 + 4xy + 16x + 8y + 20 = 0$
 2) $3x^2 + 4xy + 16x - 8y + 20 = 0$
 3) $3x^2 + 4xy - 16x - 8y + 20 = 0$
 4) $3x^2 - 4xy - 16x + 8y - 20 = 0$
65. The pair of tangents from origin to the circle $x^2 + y^2 + 4x + 2y + 3 = 0$ is
 1) $(2x + y)^2 = 3(x^2 + y^2)$
 2) $(4x + 2y)^2 = 3(x^2 + y^2)$
 3) $(2x - y)^2 = 3(x^2 + y^2)$
 4) not existing

CIRCLES-RELATIVE POSITIONS

66. The circles $x^2 + y^2 - 12x + 8y + 48 = 0$, $x^2 + y^2 - 4x + 2y - 4 = 0$ are
 1) intersecting
 2) touching externally
 3) touching internally
 4) one is lying inside the other
67. The circles $x^2 + y^2 - 2x - 4y - 20 = 0$, $x^2 + y^2 + 4x - 2y + 4 = 0$ are
 1) one lies outside the other
 2) one lies completely inside the other
 3) touch externally
 4) touch internally
68. The number of common tangents to $x^2 + y^2 = 25$, $(x - 3)^2 + (y - 4)^2 = 121$ is
 1) one 2) two
 3) four 4) zero
69. The internal centre of similitude of the circles $x^2 + y^2 - 2x + 4y + 4 = 0$, $x^2 + y^2 + 4x - 2y + 1 = 0$ divides the segment joining their centres in the ratio
 1) 1:2 2) 2:1
 3) -1:2 4) -2:1
70. The external centre of similitude of the circle $x^2 + y^2 - 12x + 8y + 48 = 0$ and $x^2 + y^2 - 4x + 2y - 4 = 0$ divides the segment joining centres in the ratio
 1) 2:3 2) 3:2
 3) -2:3 4) -3:2
71. If the two circles $(x - 1)^2 + (y - 3)^2 = r^2$ and $x^2 + y^2 - 8x + 2y + 8 = 0$ intersect in two distinct points, then (AIEEE-2003)
 1) $r > 2$ 2) $2 < r < 8$
 3) $r < 2$ 4) $r = 2$
72. If the distance between the centres of two circles of radii 3, 4 is 25 then the length of the transverse common tangent is
 1) 24 2) 12
 3) 26 4) 13

LEVEL - I (C.W.) - KEY

- 1) 2 2) 2 3) 1 4) 1 5) 1 6) 2 7) 4
 8) 3 9) 2 10) 1 11) 3 12) 2 13) 1 14) 4
 15) 2 16) 4 17) 1 18) 2 19) 1 20) 3 21) 3
 22) 2 23) 1 24) 4 25) 2 26) 4 27) 3 28) 4
 29) 2 30) 4 31) 1 32) 2 33) 2 34) 4 35) 3
 36) 2 37) 1 38) 4 39) 3 40) 1 41) 3 42) 1
 43) 3 44) 3 45) 1 46) 1 47) 1 48) 1 49) 1
 50) 1 51) 2 52) 3 53) 2 54) 3 55) 3 56) 1
 57) 2 58) 1 59) 2 60) 2 61) 4 62) 1 63) 3
 64) 3 65) 1 66) 2 67) 2 68) 1 69) 1 70) 3
 71) 2 72) 1

LEVEL - I (C.W.) - HINTS

- 1) Point of intersection of straight lines is centre
 2) same centre
 3) Let the required circle is $x^2 + y^2 - 6x - 4y + k = 0$
 and it is passing through $(-2, 14)$
 4) G is centre and radius is AG
 5) $a = 1$
 6) Point of intersection of normals is centre
 7) radius is $2a$
 8) $r = CP$
 9) $r_1 = r_2$ and $(g, f) = (x, y)$
 10) verify
 11) $CP - r$
 12) perpendicular distance - radius
 13) circle is $x^2 + y^2 - ax - by = 0$ where
 $a = 2; b = -2$
 14) $x^2 + x^2 = (2r)^2$
 15) The lines form a square
 16) $a_1 a_2 = b_1 b_2$
 17) split lines and center is passing through point of intersection of midway lines
 18) $S_{11} = 3$
 19) $C = (3, 2)$, $r = S_{11}$
 20) it is power of point $(3, 4)$
 21) $2\sqrt{r^2 - d^2} = 24$; $d = 5$
 22) $r > d$
 23) Divide with 2 and apply $\sqrt{S_{11}}$

- 24) Let point be (x_1, y_1) and $\sqrt{S_{11}} = 3$
 25) $S_1 = 0$
 26) $y = mx \pm r\sqrt{1+m^2}$
 27) $r = d$
 28) $S_1^2 = SS_{11}$
 29) Passing through the origin
 30) $y = mx \pm r\sqrt{1+m^2}$ and substitute $(7, 1)$
 31) use $\tan \frac{\theta}{2} = \frac{r}{\sqrt{S_{11}}}$
 32) Director Circle
 33) $x^2 + y^2 = a^2 + b^2$
 34) director circle concept
 35) verify
 36) $\frac{\sqrt{S_{11}}}{\sqrt{S_{11}^1}} = \frac{2}{3}$
 37) verify
 38) $r = d$
 39) verify
 40) $2\sqrt{f^2 - c}$
 41) $r = 6, d = 2$ and $2\sqrt{r^2 - d^2} = 8\sqrt{2}$
 42) Verify $g^2 - f^2 = 12$
 43) Lines are two parallel tangents. Centre lies on diameter
 44) foot of the perpendicular
 45) verify
 46) verify
 47) draw the diagram
 48) $x = r \cos \theta$; $y = r \sin \theta$
 49) eliminate θ
 50) Apply $(x + g)\cos\theta + (y + f)\sin\theta - r = 0$
 51) $S_1 = 0$
 52) $S_1 = 0$
 53) $S_1 = 0$ and compare
 54) $S_1 = 0$ and compare
 55) $S_{12} = 0$
 56) $r^2 (l_1 l_2 + m_1 m_2) = n_1 n_2$

- 57) $S_{12} = 0$; $CP.CQ \neq r^2$
 58) $S_1 = 0$ and applying $2\sqrt{r^2 - d^2}$
 59) find inverse point and compare
 60) verify
 61) foot of the perpendicular to its polar
 62) line passing through two points
 63) $S_1 = S_{11}$
 64) $S_1^2 = S.S_{11}$
 65) $S_1^2 = S.S_{11}$
 66) $C_1C_2 = r_1 + r_2$
 67) $C_1C_2 < |r_1 - r_2|$
 68) $C_1C_2 = |r_1 - r_2|$
 69) $r_1 : r_2$ internally
 70) $r_1 : r_2$ externally
 71) $C_1C_2 < r_1 + r_2$
 72) $\sqrt{d^2 - (r_1 + r_2)^2}$



LEVEL-I (H.W)

EQUATION OF CIRCLE, CENTRE - RADIUS FORM

1. The circle with centre $\left(\frac{3}{2}, \frac{1}{2}\right)$ and radius $\sqrt{\frac{3}{2}}$ is
 1) $x^2 + y^2 - 2x - 2y + 1 = 0$
 2) $x^2 + y^2 + 2x + 2y + 1 = 0$
 3) $x^2 + y^2 - 3x - y + 1 = 0$
 4) $x^2 + y^2 + 3x + y - 1 = 0$
 2. Circle with centre origin and passing through $(-1, 2)$ is
 1) $x^2 + y^2 = 5$
 2) $x^2 + y^2 = 1$
 3) $x^2 + y^2 = 2$
 4) $x^2 + y^2 = 4$

3. Equation of circle passing through $(-1, -2)$ and concentric with the circle

$$x^2 + y^2 + 3x + 4y + 1 = 0$$

- 1) $x^2 + y^2 + 3x + 4y + 6 = 0$
 2) $x^2 + y^2 - 3x - 4y + 6 = 0$
 3) $x^2 + y^2 + 3x + 4y - 6 = 0$
 4) $x^2 + y^2 + 3x - 4y - 6 = 0$

4. The area of the circle

$x^2 + y^2 - 4x - 2y + k = 0$ is 25π square units then $k =$

- 1) 20 2) -20
 3) ± 20 4) 0

5. If $(1, 2)$ and $(2, a)$ are extremities of a diameter of the circle $x^2 + y^2 - 3x - 4y + 6 = 0$ then $a =$

- 1) 1 2) 2
 3) 3 4) 4

6. The lines $2x - 3y = 5$ and $3x - 4y = 7$ are the diameters of a circle of area 154 sq. units. Then equation of circle is

- 1) $(x+1)^2 + (y+1)^2 = 49$
 2) $(x-1)^2 + (y-1)^2 = 49$
 3) $(x-1)^2 + (y+1)^2 = 49$
 4) $(x+1)^2 + (y-1)^2 = 49$

SHORTEST DISTANCE AND LONGEST DISTANCE

7. The nearest point on the circle

$x^2 + y^2 - 6x + 4y - 12 = 0$ from $(-5, 4)$ is

- 1) $(1, 1)$ 2) $(-1, 1)$
 3) $(-1, 2)$ 4) $(-2, 2)$

8. If d_1 & d_2 are the longest and shortest distances of $(-7, 2)$ from any point (α, β) on the curve whose equation is $x^2 + y^2 - 10x - 14y = 51$ then G.M of d_1 & d_2 is

- 1) $\sqrt{11}$ 2) 7
 3) 2 4) $2\sqrt{11}$

CONCYCLIC POINTS, CIRCUMSCRIBING CIRCLES

9. The circle through the points (2,3), (2,2), (3,2) is

1) $x^2 + y^2 + 2x + 3y = 0$ 2) $x^2 + y^2 = 13$

3) $x^2 + y^2 - 5x - 5y + 12 = 0$

4) $x^2 + y^2 + 5x + 5y + 12 = 0$

10. The circle passing through (5,-5), (1,7), (-7,1) is

1) $x^2 + y^2 = 50$ 2) $x^2 + y^2 - 5x + 5y = 0$

3) $x^2 + y^2 - 6x - 8y = 0$ 4) $x^2 + y^2 = 5$

11. Centre of circle passing through A(0,1), B(2,3), C(-2,5) is

1) (-1, 10) 2) $\left(-\frac{1}{3}, \frac{10}{3}\right)$

3) $\left(\frac{10}{3}, -\frac{2}{3}\right)$ 4) $\left(\frac{1}{3}, \frac{10}{3}\right)$

POWER OF A POINT, POSITION OF A POINT, CHORD

12. A chord through P cut the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ in A and B another chord through P in C and D, then

1) PA.PB < PC.PD 2) PA.PB = PC.PD

3) PA.PC = PB.PD 4) PA.PB > PC.PD

13. The intercept made by the circle

$x^2 + y^2 - 4x - 6y - 3 = 0$ on the line $x + y - 3 = 0$ is

1) $\sqrt{14}$ 2) $2\sqrt{14}$ 3) $6\sqrt{2}$ 4) $3\sqrt{2}$

TANGENT AND NORMAL

14. The equation of the tangent to the circle

$x^2 + y^2 - 4x + 4y - 2 = 0$ at (1,1) is

1) $x - 3y + 2 = 0$ 2) $x + 3y - 2 = 0$

3) $3x + y - 1 = 0$ 4) $x + 3y = 4$

15. The equations of the tangents to the circle $x^2 + y^2 = 4$ which are parallel to x-axis are

1) $x = \pm 2$ 2) $y = \pm 2$

3) $y = \pm 1$ 4) $y = \pm 4$

16. The point of contact of $y = x + 3\sqrt{2}$ with $x^2 + y^2 = 9$ is

1) $\left(\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$ 2) $\left(\frac{3}{\sqrt{2}}, \frac{2}{\sqrt{2}}\right)$

3) $\left(\frac{-3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$ 4) $\left(\frac{3}{\sqrt{2}}, \frac{-3}{\sqrt{2}}\right)$

17. If the lines $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are tangents to a circle, then the radius of the circle is

1) $\frac{3}{4}$ 2) $\frac{1}{2}$ 3) $\frac{7}{8}$ 4) $\frac{11}{10}$

18. Equation of the tangent to the circle $x^2 + y^2 - 2x + 4y - 4 = 0$ which is parallel to the line $3x + 4y - 1 = 0$ is

1) $3x + 4y = 5$ 2) $3x + 4y = 15$

3) $3x + 4y = 10$ 4) $3x - 4y = 10$

19. The tangents at (5,12) and (12,-5) to the circle $x^2 + y^2 = 169$ are

1) coincident 2) perpendicular

3) parallel 4) at an angle of 45°

20. If two tangents are drawn from a point on $x^2 + y^2 = 16$ to the circle $x^2 + y^2 = 8$ then the angle between the tangents is

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{4}$ 3) $\frac{2\pi}{3}$ 4) π

21. The length of tangent from any point on the circle $x^2 + y^2 + 4x - 6y - 12 = 0$ to the circle $x^2 + y^2 + 4x - 6y + 4 = 0$ is

1) 2 2) 16 3) 8 4) 4

CIRCLES TOUCHING COORDINATE AXES AND INTERCEPTS ON AXES

22. The number of circles touching both the axes with radius 5 is

1) 4 2) 3 3) 2 4) 0

23. The intercept made by the circle $x^2 + y^2 + 4x - 8y + c = 0$ on x-axis is $2\sqrt{10}$, then c =

1) -6 2) 6

3) ± 6 4) 12

24. The equation to the side BC of $\triangle ABC$ is $x + 5 = 0$. If $(-3, 2)$ is the orthocentre and $(0, 0)$ is the circumcentre then radius of the circle is

- 1) $\sqrt{51}$ 2) $\sqrt{13}$ 3) $\sqrt{53}$ 4) $\sqrt{63}$

25. Equation of the circle with centre $(5, -4)$ and touching y-axis is

- 1) $x^2 + y^2 - 10x + 8y + 16 = 0$
 2) $x^2 + y^2 + 10x + 8y + 16 = 0$
 3) $x^2 + y^2 - 10x - 8y + 16 = 0$
 4) $x^2 + y^2 + 10x - 8y + 16 = 0$

PARAMETRIC EQUATIONS

26. The parametric equations of the circle $x^2 + y^2 + 2x + 4y - 11 = 0$ are

- 1) $x = 1 + 4\cos\theta, y = 2 + 4\sin\theta$
 2) $x = -1 + 4\cos\theta, y = 2 + 4\sin\theta$
 3) $x = -1 + 4\cos\theta, y = -2 + 4\sin\theta$
 4) $x = 1 - 4\cos\theta, y = 2 - 4\sin\theta$

27. Locus of the point $(\cos\theta + \sin\theta, \cos\theta - \sin\theta)$

Where θ is parameter is

- 1) $x^2 + y^2 = 1$ 2) $x^2 + y^2 = 4$
 3) $x^2 + y^2 = 2$ 4) $y^2 = 4ax$

28. The normal at ' θ ' of the circle $x^2 + y^2 = a^2$ is

- 1) $x\cos\theta + y\sin\theta = 0$ 2) $x\cos\theta - y\sin\theta = 0$
 3) $x\sin\theta + y\cos\theta = 0$ 4) $x\sin\theta - y\cos\theta = 0$

CHORD OF CONTACT, POLE, POLAR, CONJUGATE POINTS AND LINES, INVERSE POINTS (POLE AND POLAR NOT FOR MAINS)

29. The chord of contact of $(1, 2)$ with respect to the circle $x^2 + y^2 - 4x - 6y + 2 = 0$ is

- 1) $x + y - 6 = 0$ 2) $x + 2y - 2 = 0$
 3) $2x + y + 6 = 0$ 4) not existing

30. The polar of $(2, 3)$ w.r.t the circle $x^2 + y^2 - 4x - 6y + 2 = 0$ is

- 1) a tangent 2) a diameter
 3) a chord of contact 4) not existing

31. If $(1, 4), (-2, 3)$ are conjugate points w.r.t $x^2 + y^2 = k$, then $k =$

- 1) 10 2) $\sqrt{10}$ 3) 100 4) 4

32. If $kx + 3y = 1, 2x + y + 5 = 0$ are conjugate lines w.r.t the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ then $k =$

- 1) 3 2) 4 3) 2 4) 1

33. The line $x\cos\alpha + y\sin\alpha = p$ intersects the circle $x^2 + y^2 = 4$ at A and B. If the chord AB makes an angle 30° at a point on the circumference of the circle, then

- 1) $p^2 = 3$ 2) $p^2 = 4\cos^2 15^\circ$
 3) $p^2 = 2$ 4) $p^2 = 6$

34. The lengths of tangents from two points A, B to a circle are 6, 7 respectively. If A and B are conjugate points then $AB =$

- 1) 85 2) 42 3) $\sqrt{85}$ 4) $\sqrt{13}$

35. If the inverse of $P(1, 2)$ w.r.t to

$x^2 + y^2 - 4x - 6y + 9 = 0$ is Q then $PQ =$

- 1) 2 2) $\sqrt{2}$ 3) 3 4) 4

CHORD WITH MID POINT

36. The mid point of the chord $x - 2y + 7 = 0$ w.r.t the circle $x^2 + y^2 - 2x - 10y + 1 = 0$ is

- 1) $(7, 21)$ 2) $\left(\frac{7}{4}, \frac{21}{4}\right)$
 3) $\left(\frac{7}{5}, \frac{21}{5}\right)$ 4) $\left(\frac{-7}{5}, \frac{-21}{5}\right)$

CIRCLES-RELATIVE POSITIONS

37. The circles $x^2 + y^2 - 8x + 6y + 21 = 0$, $x^2 + y^2 + 4x - 10y - 115 = 0$ are

- 1) intersecting 2) touching externally
 3) touching internally 4) one is lying inside the other

38. The number of common tangents to $x^2 + y^2 = 8$, $x^2 + y^2 = 2$ is

- 1) one 2) two
 3) four 4) zero

39. The internal centre of similitude of two circles

$(x - 3)^2 + (y - 2)^2 = 9, (x + 5)^2 + (y + 6)^2 = 9$ is

- 1) $(-1, -2)$ 2) $(-2, -1)$
 3) $(3, 2)$ 4) $(-5, -6)$

40. If the circles $x^2 + y^2 + 2gx + 2fy = 0$
 $x^2 + y^2 + 2g^1x + 2f^1y = 0$ touch each other
 then

- 1) $fg = f^1g^1$ 2) $fg^1 = f^1g$
 3) $f + g = f^1 + g^1$ 4) $f + f^1 = g + g^1$

41. Equation of circle whose radius is 5 and which
 touches the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ at
 (5,5) is

- 1) $(x-9)^2 + (y-8)^2 = 5$
 2) $(x-9)^2 + (y+8)^2 = 25$
 3) $x^2 + y^2 = 25$
 4) $(x-9)^2 + (y-8)^2 = 25$

LEVEL-I (H.W.) KEY

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1) 3 | 2) 1 | 3) 1 | 4) 2 | 5) 2 | 6) 3 | 7) 2 |
| 8) 4 | 9) 3 | 10) 1 | 11) 2 | 12) 2 | 13) 2 | 14) 1 |
| 15) 2 | 16) 3 | 17) 1 | 18) 3 | 19) 2 | 20) 1 | 21) 4 |
| 22) 1 | 23) 1 | 24) 3 | 25) 1 | 26) 3 | 27) 3 | 28) 4 |
| 29) 4 | 30) 4 | 31) 1 | 32) 3 | 33) 1 | 34) 3 | 35) 2 |
| 36) 3 | 37) 3 | 38) 4 | 39) 1 | 40) 2 | 41) 4 | |

LEVEL-I (H.W.) HINTS

- $(x-h)^2 + (y-k)^2 = r^2$
- $r = CP$
- same centre
- $(radius)^2 = 25$
- $2 + 2a = 6$
- centre is P.I and $\pi r^2 = 154$
- verify
- GM = square root of 2 numbers
- verify
- verify
- SA = SB = SC
- $PA.PB = S_{11}$ & $PC.PD = S_{11}$
- $2\sqrt{r^2 - d^2}$

- $S_1 = 0$
- $m=0$
- Distance is diameter
- $r = d$
- $m_1m_2 = -1$
- perpendicular
- $\sqrt{|c_1 - c_2|}$
- four circles in four quadrants
- $2\sqrt{g^2 - c}$
- Let D is the image of (-3,2) w.r.t $x+5=0$. And $r=SD$
- verify
- $x = h + r \cos \theta; y = k + r \sin \theta$ where (h, k) is centre
- squaring and adding
- verify
- $S_{11} < 0$
- $S_1 = 0$
- $S_{12} = 0$
- apply formula
- $\angle AOB = 2\angle APB = 60^\circ$
Homogenous
- $\sqrt{a^2 + b^2}$ where $a = 6$ and $b = 7$
- find inverse point and distance
- $S_1 = S_{11}$
- $C_1C_2 = |r_1 - r_2|$
- concentric circles
- find point divides $r_1 : r_2$ internally
- C_1, O, C_2 are collinear
- verify